



FULL LENGTH MOCK TEST #04 (BEGINNERS)

Time: 90mins

Marks: 300

[1]: Arrange in descending order of the units of memory TB, KB, GB, MB.

Option 1:

TB > MB > GB > KB

Option 3:

TB > GB > MB > KB

Option 2:

MB > GB > TB > KB

Option 4:

GB > MB > KB > TB

[2]: Which of the following functionalities of an operating system allocates memory space to different programs under execution?

Option 1:

Security Management

Option 3:

Memory Management

Option 2:

Processor Management

Option 4:

Device Management

[3]: _____ is a small program that switches the processor from one process to another.

Option 1:

Scheduler

Option 2:

Dispatcher

Option 3:

Swapper

Option 4:

Lazy swapper

[4]: Which of the following are the techniques of

implementing virtual memory?

Option 1:

Overlays

Option 2:

Paged Virtual Memory

Option 3:

Segmented Virtual Memory

Option 4:

All of the above

[5]: Which of the following operating systems do NOT / does NOT support multitasking?

(i) Windows

(ii) Linux

(iii) MS-DOS

Option 1:

Both (i) and (iii)

Option 2:

Only (ii)

Option 3:

Only (iii)

Option 4:

Both (ii) and (iii)

[6]: In Kernel I/O Subsystem, A _____ is a buffer that holds jobs for a device until it is ready to take them.

Option 1:

Buffering

Option 2:

Caching

Option 3:

Spooling

Option 4:

Scheduling

[7]: Kernel is the core part of _____.

Option 1:

Linux Operating System

Option 2:

Pytorch Software

Option 3:

Microsoft Dotnet

Option 4:

Java

[8]: Consider a process of 200 KB which has 2 segments each segment size is 256 KB. Which of the following is the correct number of bytes of overhead in segmented paging if the page table entry is 4B, the segment table entry is 2B and Page size is 2KB.

Option 1:

512 B

Option 2:

516 B

Option 3:

256 B

Option 4:

260 B

[9]: Consider the following four processes with the arrival time and length of CPU burst given in milliseconds :

Process	Arrival Time	Burst Time
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

The average waiting time for **preemptive SJF scheduling** algorithm is

Option 1:

6.5 ms

Option 2:

7.5 ms

Option 3:

6.75 ms

Option 4:

7.75 ms

[10]: An operating system uses a Preemptive Priority scheduling algorithm. Consider the arrival time, priority (0 is the lowest) and CPU burst, I/O burst, another CPU burst. Assume that each process has its own I/O resource.

			Burst time		
Process	Arrival Time	Priority	CPU I/O	CPU	CPU
P ₁	0	1	2	4	1
P ₂	1	2	2	5	3
P ₃	3	3	3	5	2

What is the finishing time of process P1?

Option 1:

11

Option 2:

13

Option 3:

14

Option 4:

15

[11]: Multi-threading on a multi-CPU machine

Option 1:

decreases concurrency

Option 2:

increases concurrency

Option 3:

doesn't affect the concurrency

Option 4:

can increase or decrease the concurrency

[12]: Consider an operating system containing 4 processes and each process requires 3 instances of resource M. What is the minimum number of instances of resource M is needed such that no deadlock will ever occur?

Option 1:

8

Option 2:

9

Option 3:

10

Option 4:

11

[13]: A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is

Option 1:	Option 2:	Option 3:	Option 4:
11 bits	13 bits	15 bits	20 bits

[14]: Consider a counting semaphore which was initialized to $(y + 7)$ for a system. Seven wait and $(y + 3)$ signal operations were completed on this semaphore. If the final value of the semaphore is $(y + 2)$ then what is the value of y ?

Option 1:	Option 2:	Option 3:	Option 4:
-2	-1	0	1

[15]: Which of the following is not an example of secondary storage device ?

Option 1:	Option 2:	Option 3:	Option 4:
R.A.M	Hard disks	Magnetic tapes	C.D.

[16]: The device which is used to carry digital data on analogue lines is called as?

Option 1:	Option 2:	Option 3:	Option 4:
Multiplexer	Modulator	Demodulator	Modem

[17]: A flip-flop has _____.

Option 1:

One stable state

Option 2:

No stable state

Option 3:

Two stable state

Option 4:

None of the above

[18]: How to implement D flip-flop using J-K flip-flop?

Option 1:

$J = 0, K = D$

Option 2:

$J = K = D$

Option 3:

$J = 1, K = D$

Option 4:

$J = D, K = \bar{D}$

[19]: The minimum number of flip-flops needed for a divide by 42 ripple counter is:

Option 1:

7

Option 2:

4

Option 3:

5

Option 4:

6

[20]: The following truth-table belongs to which one of the four gates-

A	B	X
1	1	0
0	1	0
1	0	0
0	0	1

Option 1:

OR

Option 2:

NAND

Option 3:

XOR

Option 4:

NOR

[21]: The Logic gates are built by using

Option 1:

Semiconductor diode

Option 2:

Conductor

Option 3:

Transistor

Option 4:

Both 1 and 3

[22]: Match the columns:

List - A	List - B
(i) XOR	(a) $Y = A \cdot B + \bar{A} \cdot \bar{B}$
(ii) XNOR	(b) $Y = \bar{A} \cdot B + A \cdot \bar{B}$
(iii) NOR	(c) $Y = \overline{A+B}$

Option 1:

(i) - (c), (ii) - (b), (iii) - (a)

Option 2:

(i) - (b), (ii) - (a), (iii) - (c)

Option 3:

(i) - (a), (ii) - (c), (iii) - (b)

Option 4:

(i) - (c), (ii) - (a), (iii) - (b)

[23]: The output of an AND gate with three inputs, A, B and C, is HIGH when _____.

Option 1:

$A = 1, B = 1, C = 0$

Option 2:

$A = 0, B = 0, C = 0$

Option 3:

$A = 1, B = 1, C = 1$

Option 4:

$A = 1, B = 0, C = 1$

[24]: The minimum number of gates required to implement

$$X = AB + ABC + A\bar{B}\bar{C} + A\bar{C}$$

(Assume complement are not available and all gates are available)

Option 1:

1

Option 2:

2

Option 3:

3

Option 4:

4

[25]: De Morgan's theorem states that

Option 1:

$$(X + Y)' = Y' + X'$$

Option 2:

$$(X \cdot Y)' = X' + Y'$$

Option 3:

$$(X \cdot Y)' = Y' \cdot X'$$

Option 4:

$$(X + Y)' = X' + Y'$$

[26]: Consider the following statements:

I. The second largest element in a max-heap is always a child of the root node

II. A max-heap can be constructed from a binary search tree in $\Theta(n)$ time

Which of the above statements is/are FALSE if heap consists of distinct elements?

Option 1:

Only I

Option 2:

Only II

Option 3:

Both I and II

Option 4:

None of these

[27]: Which type of linked list stores the address of the header node in the next field of the last node ?

Option 1:

Singly linked list

Option 2:

Circular linked list

Option 3:

Doubly linked list

Option 4:

Hashed list

[28]: The minimum number of bits required to represent $(-128)_{10}$ in 2's complement form and signed form respectively are:

Option 1:

7, 8

Option 2:

8, 8

Option 3:

8, 9

Option 4:

8, 7

[29]: Consider the equation $(146)_b + (313)_{b-2} = (246)_8$. Which of the following is the value of b?

Option 1:

8

Option 2:

7

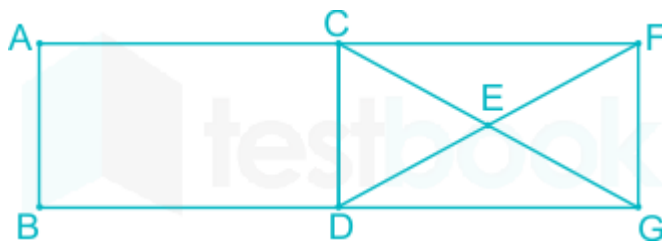
Option 3:

10

Option 4:

16

[30]: Which sequence cannot be obtained while using BFS traversal for the below given graph?



I. A B C F G E D

II. A B C G F E D

III. A B D E C F G

Option 1:

Only I

Option 2:

Only II

Option 3:

Only II and III

Option 4:

I, II, III

[31]: The recurrence relation $T(n) = 7T(n/7) + n$ has the solution:

Option 1:

$O(n)$

Option 2:

$O(\log n)$

Option 3:

$O(n \log(n))$

Option 4:

$O(n^2)$

[32]: Depth First Search graph search algorithm uses _____ data structure for its implementation.

Option 1:

Dequeue

Option 2:

Queue

Option 3:

tree

Option 4:

Stack

[33]: The maximum height of a binary tree with 'n' nodes is _____.

Option 1:

n

Option 2:

$(n - 1)$

Option 3:

$(n + 1)$

Option 4:

$(n + n)$

[34]: An m-way search tree T is a tree in which all the nodes are of degree

Option 1:

$\text{Degree} \leq m$

Option 2:

$\text{Degree} = m + 1$

Option 3:

$\text{Degree} = m$

Option 4:

$\text{Degree} > m$

[35]: In binary search tree which traversal is used for getting ascending order values?

Option 1:

Inorder

Option 2:

Preorder

Option 3:

Postorder

Option 4:

None of the options

[36]: How many triangles are there in the given figure?



Option 1:

30

Option 2:

24

Option 3:

32

Option 4:

36

[37]: Looking at a portrait of a man, Harsh said, "His mother is the wife of my father's son. Brothers and sisters I have none. "At whose portrait was Harsh looking?

Option 1:

His son

Option 2:

His cousin

Option 3:

His uncle

Option 4:

His nephew

[38]: If in a certain language A is coded as 1, B is coded as 2, and so on, how is B I D D I C coded in that code?

Option 1:

294493

Option 2:

284563

Option 3:

375582

Option 4:

394492

[39]:

What should come in place of the question mark '?' in the following number series?

2, 6, 22, ?, 65814

Option 1:

96

Option 2:

58

Option 3:

278

Option 4:

198

[40]:

Wrong number in the series 7, 8, 18, 57, 228, 1165, 6996

Option 1:

228

Option 2:

18

Option 3:

57

Option 4:

8

[41]:

Given below are two statements, Consider these statements to be true even if they seem factually absurd, Read the conclusions and then decide which of the given conclusions logically follow(s) from the given statements?

Statements :

All vases are flowers.

No flowers is a plant.

Conclusions :

I. No vases is a plant.

II. Some plant are vases

Option 1:

Only conclusion I follows.

Option 2:

Both the conclusions follow.

Option 3:

Only conclusion II follows.

Option 4:

Neither conclusion follows.

[42]:

Niranjan goes 6 km towards East from Library, and takes a right turn. Now he goes 3 km and takes a left turn. He goes 6 km and takes a right turn. He again goes 3 km and takes a right turn. He now goes 6 km and takes a left turn. He now travels 3 km and takes a left turn to reach his home which is

6 km away. What is the shortest distance between library and Niranjan's home?

Option 1:

15 km

Option 2:

17 km

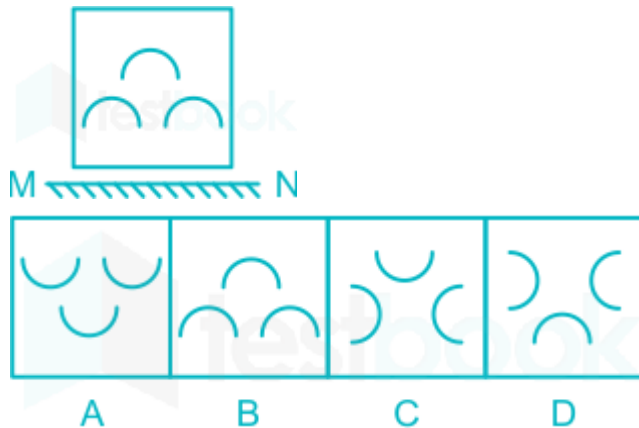
Option 3:

19 km

Option 4:

None of the above

[43]: Select the mirror image for the following figure.



Option 1:

C

Option 2:

D

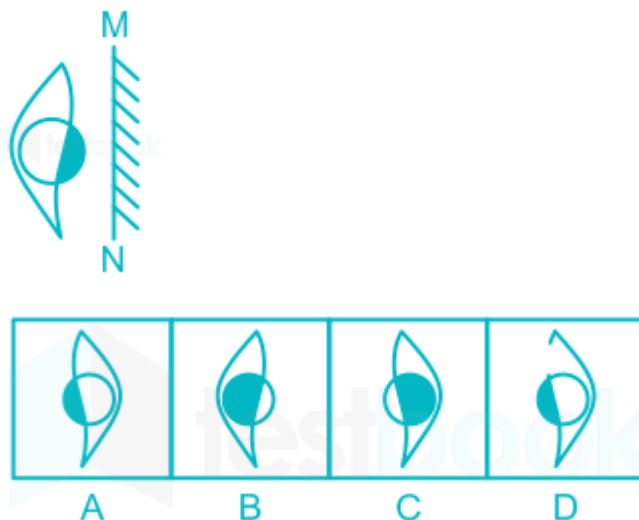
Option 3:

A

Option 4:

B

[44]: Choose the mirror image for the following figure when the mirror is kept on the MN line.



Option 1:

C

Option 2:

B

Option 3:

D

Option 4:

A

[45]: Choose the correct alternative based on relationship
Analogy 2021

Planet : Orbit :: Projectile : ?

Option 1:

Trajectory

Option 2:

Track

Option 3:

Milky Way

Option 4:

Path

[46]:

Match List I and List II **Analogy**

LIST I		LIST II	
A.	Dog : Rabies :: Mosquito :	I.	Bacteria
B.	Amnesia : Memory :: Paralysis :	II.	Liver
C.	Meningitis : Brain :: Cirrhosis :	III.	Movement
D.	Influenza : Virus :: Typhoid :	IV.	Malaria

Choose the correct answer from the options
given below:

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Option 1:

A-II, B-III, C-I, D-IV

Option 3:

A-IV, B-III, C-II, D-I

Option 2:

A-III, B-IV, C-II, D-I

Option 4:

A-IV, B-III, C-I, D-II

[47]:

The average of 50 numbers is 30. If two numbers, 35 and 40 are
discarded, then the average of the remaining numbers is nearly:

Option 1:

28.32

Option 2:

29.68

Option 3:

28.78

Option 4:

29.27

[48]: If a matrix A is symmetric as well as skew-symmetric, then

Option 1:

A is a diagonal matrix

Option 2:

A is a null matrix

Option 3:

A is a unit matrix

Option 4:

A is a triangular matrix

[49]: A and B be 3×3 matrices. Then $|A \cdot B| = 0$ implies

Option 1:

$A = 0$ or $B = 0$

Option 2:

$|A| = 0$ and $|B| = 0$

Option 3:

$|A| = 0$ or $|B| = 0$

Option 4:

$A = 0$ and $B = 0$

[50]: If P and Q are two sets, then $(P - Q) \cup (Q - P) \cup (P \cap Q)$ will be

Option 1:

P

Option 2:

Q

Option 3:

$P \cap Q$

Option 4:

$P \cup Q$

[51]: If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$; $a, b, c \in R$, have a common root, then $a:b:c$ is equal to

Option 1:

$1:2:3$

Option 2:

$3:2:1$

Option 3:

$1:3:2$

Option 4:

$3:1:2$

[52]:

If $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P. then which of the following is in A.P.?

Option 1:

a, b, c

Option 2:

a^2, b^2, c^2

Option 3:

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$

Option 4:

None of these

[53]: If roots of the equation $x^3 - 12x^2 + 39x - 28 = 0$ are in AP, then its common difference is -

(1) more than -4

(2) ± 2

(3) ± 3

(4) ± 4

Option 1:

1, 2 and 3 are correct

Option 2:

1 and 2 are correct

Option 3:

2 and 4 are correct

Option 4:

1 and 3 are correct

[54]: The number of ways in which 7 plus (+) signs and 5 minus (-) signs can be arranged in a row so that no two minus signs are together, is

Option 1:

${}^8C_5 \times 5!$

Option 2:

${}^7C_5 \times 5!$

Option 3:

8C_5

Option 4:

$7!$

[55]:

If a and b are the greatest values of ${}^{2n}C_r$, and ${}^{2n-1}C_r$, respectively. Then,

Option 1:

$$a = 2b$$

Option 2:

$$b = 2a$$

Option 3:

$$a = b$$

Option 4:

none of these

[56]: How many diagonals are there in a polygon with n sides?

Option 1:

$$\frac{n(n-1)}{2}$$

Option 2:

$$\frac{n(n-3)}{2}$$

Option 3:

$$\frac{n(n+1)}{2}$$

Option 4:

none

[57]: Find the equation of the ellipse whose axes are along the coordinate axes, vertices are $(\pm 5, 0)$ and foci at $(\pm 4, 0)$.

Option 1:

$$\frac{x^2}{5} + \frac{y^2}{9} = 1$$

Option 2:

$$\frac{x^2}{25} + \frac{y^2}{3} = 1$$

Option 3:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

Option 4:

$$\frac{x^2}{15} + \frac{y^2}{12} = 1$$

[58]: The foci of the ellipse $25x^2 + 36y^2 = 225$ are

Option 1:

$$\left(\pm \frac{1}{2}\sqrt{11}, 0\right)$$

Option 2:

$$\left(\pm \frac{5}{2}, 0\right)$$

Option 3:

$$\left(0, \pm \frac{1}{2}\sqrt{11}\right)$$

Option 4:

$$\left(0, \pm \frac{5}{2}\right)$$

[59]: For any two events A and B, the probability that at least one of them occur is 0.6. If A and B occur simultaneously with a probability 0.3, then $P(A') + P(B')$ is

Option 1:

0.9

Option 2:

1.15

Option 3:

1.1

Option 4:

1.0

[60]: If A and B are two independent events with $P(A) = \frac{3}{5}$ and $P(B) = \frac{4}{9}$, then $P(A \cap B)$ equals:

Option 1:

$\frac{4}{15}$

Option 2:

$\frac{8}{45}$

Option 3:

$\frac{1}{3}$

Option 4:

$\frac{7}{12}$

[61]:

If $f(x) = \sqrt{\frac{x - \sin x}{x + \cos^2 x}}$, then $\lim_{x \rightarrow \infty} f(x)$ is

Option 1:

0

Option 2:

∞

Option 3:

1

Option 4:

none of these

[62]:

The value of $\lim_{x \rightarrow \infty} \left\{ \frac{a_1^{\frac{1}{x}} + a_2^{\frac{1}{x}} + \dots + a_n^{\frac{1}{x}}}{n} \right\}^{nx}$, is

Option 1:

$a_1 + a_2 + \dots + a_n$

Option 2:

$e^{a_1 + a_2 + \dots + a_n}$

Option 3:

$\frac{a_1 + a_2 + \dots + a_n}{n}$

Option 4:

$$a_1 a_2 \dots a_n$$

[63]:

The value of $\int_0^{\pi/4} \log(1 + \tan x) dx$ is equal to:

Option 1:

$$\frac{\pi}{8} \log_e 2$$

Option 2:

$$\frac{\pi}{4} \log_2 e$$

Option 3:

$$\frac{\pi}{4} \log_e 2$$

Option 4:

$$\frac{\pi}{8} \log_e \left(\frac{1}{2} \right)$$

[64]:

Statement I The value of the integral

$$\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}}$$

is equal to $\pi/6$.

Statement II $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

Option 1:

Statement I is true Statement II is true;
Statement II is a correct explanation of
Statement I

Option 2:

Statement I is true, Statement II is true;
Statement II is not a correct explanation of
Statement I

Option 3:

Statement I is true, Statement II is false

Option 4:

Statement I is false, Statement II is true

[65]: The line $3x + 4y = 12$ cuts the axes at A and B. if O is the origin, then area of ΔOAB is:

Option 1:

12 sq. units

Option 2:

6 sq. units

Option 3:

9 sq. units

Option 4:

none

[66]:

If A is the null set, then the number of elements in the power set $P(P(\phi))$ is

Option 1:

0

Option 2:

2

Option 3:

1

Option 4:

none of these

[67]: If A and B are two events such that $P(A \cup B) = P(A \cap B)$, then the incorrect statement amongst the following statements is:

Option 1:

A and B are equally likely

Option 2:

$P(A \cap B') = 0$

Option 3:

$P(A' \cup B) = 0$

Option 4:

$P(A) + P(B) = 1$

[68]: The boolean expression

$$Z = \overline{A}\overline{B}\overline{C}D + \overline{A}BC\overline{D} + A\overline{B}\overline{C}D + AB\overline{C}\overline{D}$$

can be minimized to?

Option 1:

$$Z = \overline{A}BC + \overline{B}\overline{C}D + AB\overline{C}\overline{D}$$

Option 2:

$$Z = \overline{A}BC\overline{D} + \overline{B}\overline{C}D + AB\overline{C}\overline{D}$$

Option 3:

$$Z = \overline{A}BC\overline{D} + \overline{B}D + B\overline{C}\overline{D}$$

Option 4:

$$Z = \overline{A}BC\overline{D} + \overline{B}\overline{C}D + AB$$

[69]: Find the area of the region bounded by the curves $y = x^3$, the line $x = 2$, $x = 5$ and the x - axis?

Option 1:

173.50

Option 2:

230.25

Option 3:

175.35

Option 4:

152.25

[70]: A flip-flop circuit can be used for:

Option 1: counting Option 2: scaling Option 3: demodulation Option 4: both (1) and (2)

[71]: Match all items in Group 1 with the best match from the options given in Group 2.

Group 1	Group 2
P. Min-and max-heaps	1. File system implementation
Q. B-tress	2. Priority queue implementations
R. Arrays	3. Expensive insertion and deletion operations
S. Graphs	4. Electronic circuit design analysis

Option 1: P - 2, Q - 3, R - 4, S - 1 Option 2: P - 4, Q - 3, R - 2, S - 1
Option 3: P - 2, Q - 1, R - 3, S - 4 Option 4: P - 3, Q - 4, R - 1, S - 2

[72]: What is the focus of the parabola $x^2 = 16y$?

Option 1: (4, 0) Option 2: (0, 4) Option 3: (0, -4) Option 4: (4, 4)

[73]: $f(x) = x^x$ has a stationary point at

Option 1: $x = e$ Option 2: $x = \frac{1}{e}$ Option 3: $x = 1$ Option 4: $x = \sqrt{e}$

[74]: Box A contains 2 white and 3 red balls and box B contains 4 white and 5 red balls. One ball is drawn at random from one of the boxes and is found to

be red. Then, the probability that it was from box B, is

Option 1:

$$\frac{25}{52}$$

Option 2:

$$\frac{21}{52}$$

Option 3:

$$\frac{7}{52}$$

Option 4:

$$\frac{15}{52}$$

[75]: Select the option that is related to the third word in the same way as the second word is related to the first word. (The words must be considered as meaningful English words and must not be related to each other based on the number of letters/number of consonants/vowels in the word)
Botany : Plants : : Conchology : ?

Option 1:

Names

Option 2:

Shells

Option 3:

Soil]

Option 4:

Fossils